



SPA120 User Manual

Dual Source Picoammeter

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1	Notices	7
1.1	Copyright	7
1.2	Part Number	7
1.3	Issue	7
1.4	Location	7
1.5	Published By	7
1.6	Notes	8
1.7	Trademark Acknowledgement	8
1.8	Purpose of Manual	8
1.9	Safety Warnings	8
1.10	REACH / CLP Compliance Statement	8
2	Getting Started	9
2.1	What is the SPA120?	9
2.2	Key Specifications	9
2.3	System Requirements	10
2.4	What is in the Box	10
3	Instrument Indicators	10
4	Quick Start	10
5	Introduction	11
5.1	Welcome	11
5.2	Before You Start	11
5.3	Safety and Handling	11
5.4	Maximum Ratings	12
5.5	Electrostatic Discharge (ESD) Precautions	12
5.6	Package Contents	12
5.7	Initial Inspection	12
5.8	Measurement Best Practices	13
5.9	Operating Environment	13
5.10	What You Will Need	13
5.11	Installation Sequence	13

5.12	Installing EPIC Software	14
5.13	Installing the USB Driver	14
6	Connecting Your Instrument	14
6.1	USB Connection	15
6.2	COM Port Selection	15
6.3	Connection Status	15
6.4	Troubleshooting Connection Issues	15
7	Input Settings	16
7.1	Range Selection	16
7.2	Sample Rate	16
7.3	Null / Zero	17
7.4	Rolling Averaging	17
8	Source Supply	18
8.1	Voltage Control	18
8.2	Polarity	19
8.3	Source On/Off	19
9	Graph Display	19
9.1	Main Graph	19
9.2	Graph Title	19
9.3	Graph Key	20
9.4	Sub-Graph Overview	20
9.5	Timebase	20
9.6	Y-Axis Scale	21
9.7	Y-Axis Mode	21
9.8	Recording	21
9.9	Pause and Resume	21
9.9.1	Pause Markers	21
9.9.2	Range Change Warning	22
10	Session Browser	22
10.1	Browsing Sessions	22
10.2	Opening a Specific Session	22
10.3	Returning to Live Data	22

11 Results Display	22
11.1 Live Current	23
11.2 Max / Min Tracking	23
11.3 Resistance Calculation	23
12 Cursors	23
12.1 X Cursors	23
12.2 Y Cursors	24
12.3 Slope Measurement	24
12.4 Cursor Results	24
13 Dual Channel Features	24
13.1 Channel Operation	25
13.2 Dual Channel Display	25
13.3 Typical Applications for Dual Channel Operation	25
14 CSV Export	25
14.1 Saving CSV Data	26
14.2 CSV File Format	26
14.3 Configuration Options	26
14.4 Auto-Naming	26
15 PNG Capture	27
15.1 Saving a PNG	27
15.2 Configuration Options	27
15.3 Auto-Naming	27
16 Metadata Overlay	27
16.1 Displayed Information	28
16.2 Draw Order	28
17 Data Streaming	28
17.1 Overview	28
17.2 CSV Streaming	29
17.2.1 CSV File Format	29
17.2.2 CSV Streaming vs Standard CSV Export	30
17.3 HTTP POST	30
17.3.1 JSON Payload Format	30

17.3.2 Authentication	31
17.3.3 HTTP POST runs on a background thread	31
17.4 Testing HTTP POST with webhook.site	31
17.4.1 Step-by-step	31
17.4.2 Alternative test endpoints	32
17.5 Streaming Interval	32
18 Gang Range and Gang Source (SPA120 Only)	33
18.1 Gang Range	33
18.2 Gang Source	33
18.3 Disabling Gang	34
19 Save / Load Setups	34
19.1 Saving a Setup	34
19.2 Loading a Setup	34
20 Settings	35
20.1 Language Tab	35
20.2 Instrument Tab	35
20.3 Comms Tab	35
20.4 File Export Tab	35
20.5 Graph Tab	36
20.6 Cursors Tab	36
20.7 Calibration Tab	36
20.8 Colours Tab	36
20.9 Setups Tab	36
20.10Data Streaming Tab	36
21 Calibration	37
22 Advanced Panel (Smorgasbord)	37
22.1 Export Functions	37
22.2 Cursor/Masurement Functions	37
22.3 Graph Functions	37
22.4 Miscellaneous	38
23 Safety Information	38

24 Troubleshooting	38
24.1 Connection Problems	38
24.2 Measurement Problems	39
24.3 Display Issues	39
24.4 File Export Issues	40
24.5 Data Streaming Issues	40
25 Support & Contact	40

1 Notices

1.1 Copyright

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The latest version of this document can be found on our website:

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Hereafter referred to as **Electron Plus**.

1.6 Notes

- We frequently update our manuals and add new features and improvements as they become available, please ensure that you check our website for an updated version of this document, especially if updating your **Electron Plus** software.
- We make every effort to ensure the accuracy of this manual's contents. If you find any errors, have suggestions for expanding on a feature, or feel that we can improve its contents then please contact us at support@electron.plus
- Copying or reproducing this document or any part of this document without written permission of **Electron Plus** is strictly prohibited.

1.7 Trademark Acknowledgement

Electron Plus fully recognises and acknowledges any trademark(s) of the respective trademark holder.

Windows™ is a trademark of Microsoft Corporation.

1.8 Purpose of Manual

The purpose of this manual is to enable you to safely setup, configure and operate your **Electron Plus** instrument, associated software and/or accessories.

Please pay particular attention to any section with a warning symbol.

1.9 Safety Warnings

Warnings, cautions and notes are colour coded through-out this manual. These are divided into several categories and are described below:

WARNING - Pay special attention to anything written here - this is for your safety and continued protection and is critical information!

CAUTION - Damage may occur to your equipment or any DUT (device under test).

NOTE - General text, with useful information or tips.

1.10 REACH / CLP Compliance Statement

Statement on Safety Data Sheet (SDS) for product:
SPA120 Dual Source Picoammeter

According to EU Regulation (EC) N° 1907/2006 (REACH) and Regulation (EC) N° 1272/2008 (CLP), no Safety Data Sheet is required for this product.

This product is classified as an article under REACH and:

- has no hazardous classification under CLP
- contains no Substances of Very High Concern (SVHC) in concentrations above 0.1% w/w and no intentional released under normal or reasonably foreseeable conditions of use (REACH Article 33)
- complies with Directive 2011/65/EU (RoHS) as amended by Directive (EU) 2015/863 (RoHS 3)

For safe use, please refer to the user manual and product labelling.

Should any changes occur in product composition or regulations, we will notify affected customers promptly.

Issued by:

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2 Getting Started

Welcome to the EPIC software for the **SPA120 Dual Source Picoammeter** by Electron+. This manual covers every aspect of operating the instrument and software. The SPA120 provides two independent measurement channels (Ch.1 and Ch.2), each with its own current measurement input and programmable bias voltage source.

2.1 What is the SPA120?

The SPA120 is a Dual Source Picoammeter that provides two independent measurement channels (Ch.1 and Ch.2). Each channel has its own current measurement input and programmable bias voltage source, allowing simultaneous measurement of two devices or two points on the same circuit. Both channels offer the full range of measurement capabilities, with the added benefit of dual-channel operation for comparative or differential measurements.

2.2 Key Specifications

Parameter	Specification
Current measurement ranges	2mA, 200uA, 20uA, 2uA, 200nA, 20nA, 2nA, 200pA
Number of ranges	8
Bias voltage supply	0V to +40V (or -40V)

Parameter	Specification
Polarity control	Positive / Negative
Sample rates	2Hz, 10Hz, 100Hz
Rolling averaging	1x, 2x, 4x, 8x, 10x, 16x, 20x, 32x, 40x, 64x
Channels	2
Interface	USB 2.0
Software	EPIC (free updates)

2.3 System Requirements

- Windows 10 or later
- USB 2.0 port for instrument connection
- Minimum 1440x900 display resolution (1920x1080 recommended)
- EPIC software (free download from www.electron.plus)

2.4 What is in the Box

- 1x SPA120 Dual Source Picoammeter
- 1x USB cable

3 Instrument Indicators

The SPA120 has one LED indicator:

- **GREEN LED** — Indicates the unit is connected and communicating with the EPIC software.

The source output state is shown by the **Source** button indicator in the EPIC software. There is no physical LED for source status.

4 Quick Start

1. Connect the SPA to your computer via the USB cable
2. Launch the EPIC software
3. Select **SPA120** as the active instrument (if not already selected)
4. Click the **Connect** button to establish communication
5. Select the desired current measurement range
6. Set the bias voltage and polarity as required

7. Connect your device or circuit to the input terminals
8. Enable the **Source** output by clicking the Source button
9. Click **Run** to begin recording and observe the live current on the graph

Always ensure the source output is disabled in EPIC before connecting or disconnecting devices from the input terminals. The Source button indicator must show OFF before handling the input connections.

5 Introduction

5.1 Welcome

Congratulations and thank you for purchasing an **Electron Plus** product.

Please take a few minutes to read the 'Before you start' section of this manual, especially as misusing this product can result in damage to it, your device-under-test or potentially place you in danger.

5.2 Before You Start

General

The SPA120 is a USB-powered instrument and does not require an external power supply under normal operating conditions.

Handle the input terminals and connectors with care. Do not force connections or over-tighten terminal posts.

The SPA120 can generate bias voltages up to 40V at the output terminals. Users are expected to be fully aware of the risks posed by these voltages and the precautions required to maintain safety.

If the Source button in EPIC shows ON, assume that a bias voltage is present at the output terminals and act accordingly!

5.3 Safety and Handling

WARNING - Read this section carefully before operating your Source Picoammeter

5.4 Maximum Ratings

The SPA120 has the following absolute maximum ratings that must never be exceeded:

Parameter	Maximum Rating
Bias voltage output	0V to ± 40 V DC
Input current (measurement terminal)	Do not exceed selected range full-scale
Operating temperature	0°C to 40°C
Storage temperature	-20°C to 60°C
Humidity	20% to 80% RH, non-condensing

Exceeding these limits may permanently damage the instrument and void your warranty.

5.5 Electrostatic Discharge (ESD) Precautions

The SPA120 contains sensitive electronic components that can be damaged by electrostatic discharge. When handling the instrument:

- Use an anti-static wrist strap when connecting/disconnecting measurement leads
- Store the instrument in its anti-static packaging when not in use
- Avoid touching the input connector centre pins directly
- Discharge yourself by touching a grounded metal object before handling

5.6 Package Contents

Your SPA120 package should contain:

- 1x SPA120 Dual Source Picoammeter
- 1x USB 2.0 cable

If any items are missing or damaged, please contact Electron Plus immediately at support@electron.plus

5.7 Initial Inspection

Before use, inspect your instrument for any damage that may have occurred during shipping:

1. Check the enclosure for cracks, dents, or other physical damage
2. Inspect the input connectors for bent pins or damage
3. Examine the USB connector for damage
4. Verify the instrument label shows your serial number

If you notice any damage, do not operate the instrument. Contact Electron Plus or your distributor immediately.

5.8 Measurement Best Practices

For accurate low-current measurements, follow these guidelines:

- Allow the instrument to warm up for a few minutes before precision measurements
- Use the Null/Zero function to cancel offset currents before taking critical readings
- Keep measurement cables as short as practical
- Use shielded cables for measurements below 20nA to minimise noise pickup
- Use guarded connections where possible for measurements below 2nA to reduce leakage paths
- Avoid fluorescent lighting and other sources of electromagnetic interference near the measurement area
- Ensure all connections are clean and free from oxidation or contamination
- Allow the measurement to stabilise — settling time increases for lower current ranges
- For ultra-low current measurements (200pA range), use a Faraday cage or shielded enclosure

5.9 Operating Environment

- **Temperature:** 0°C to 40°C (32°F to 104°F) operating, -20°C to 60°C storage
- **Humidity:** 20% to 80% relative humidity, non-condensing
- **Altitude:** Up to 2000m above sea level
- **Location:** Indoor use only, in a clean, dry environment
- **Ventilation:** Ensure adequate airflow around the instrument

5.10 What You Will Need

To operate the SPA120, you will require:

- A Windows PC meeting the system requirements
- Available USB 2.0 port
- EPIC software (free download from www.electron.plus/pages/software)
- CH340 USB driver (free download from www.electron.plus/pages/software)

5.11 Installation Sequence

Please install the **EPIC** software BEFORE connecting your instrument to a computer for the first time.

You do not need to remove a previous copy of **EPIC**, the new copy will overwrite the necessary existing file(s). The "settings.txt" file will only be created if it is not present.

When **EPIC** is first started, it will build a number of files (unless they are already there from a previous installation) in the installation directory.

EPIC software is specifically for the **Electron Plus** range of products including curve tracers, source picoammeters, analog signature analysers, and power analysers ONLY.

If upgrading from an earlier version of EPIC please be aware that the program EXE is named after the version (e.g. EPIC26_64.EXE). Previous desktop shortcuts may not work or may link to the previous version.

5.12 Installing EPIC Software

Electron Plus products require a USB connection to a PC running **EPIC** (our proprietary software) in order to function.

You can download the latest copy of **EPIC** for free from www.electron.plus/pages/software. **EPIC** is being constantly revised with new features, updates and bug fixes.

1. Download the installer ZIP file from the software page (e.g. Install_EPIC26006_64.ZIP)
2. Open the downloaded file (Windows will recognise the ZIP format and display the contents like a folder), then double click the EXE installer inside
3. Follow the on screen instructions to complete the installation

Before starting **EPIC**, we recommend installing the USB driver — see next section for details.

This manual MAY NOT represent the most up-to-date features and screen-shots, if something is unclear, please contact support@electron.plus and we will promptly try to assist you.

5.13 Installing the USB Driver

This product communicates with the host PC via USB using a CH340 bridge IC made by WCH.

A copy of the official WCH device driver is available from the SOFTWARE section of our website www.electron.plus/pages/software, device drivers can also be downloaded directly from WCH's website.

The driver we use with Windows machines is: CH341SER and is available as .EXE or a .ZIP

6 Connecting Your Instrument

6.1 USB Connection

The SPA connects to your computer via a USB 2.0 cable. The instrument uses a CH340 USB-to-serial bridge which creates a virtual COM port.

The SPA120 is USB-powered and does not require an external power supply for normal operation.

6.2 COM Port Selection

The software will attempt to auto-detect the SPA on startup. If auto-detection fails:

1. Open **Settings** by clicking the Settings button
2. Navigate to the **Comms** tab
3. Use the **Scan** button to search for available COM ports
4. Select the correct COM port from the dropdown list
5. Click **Set** to confirm

The SPA120 has its own COM port setting stored independently from other instruments. The assigned COM port is displayed under the instrument button on the Instrument selection tab.

6.3 Connection Status

The connection status is displayed in two places:

- **Connect button** in the top bar - shows the current connection state
- **Comms Status widget** (if enabled via the Advanced panel) - shows detailed communication statistics including TX, RX, and Errors

Status	Meaning
Connected	Instrument is communicating normally
Disconnected	No communication with instrument
Searching	Software is attempting to find the instrument
Error	Communication error detected

6.4 Troubleshooting Connection Issues

- Ensure the USB cable is firmly connected at both ends
- Check that the GREEN LED on the SPA is lit (indicates power and USB connection)
- Try a different USB port on your computer
- Check that the correct COM port is selected in Settings
- Ensure no other software is using the same COM port
- Check Windows Device Manager for the COM port listing
- Try restarting both the software and the instrument

7 Input Settings

The input settings control how the SPA measures current from the device under test. These controls are located in the left-hand ribbon of the main display.

7.1 Range Selection

The SPA provides eight current measurement ranges, each optimised for a specific decade of current. Select the range using the **Range** control in the left ribbon. Use the + and - buttons to step through the available ranges.

Range	Full Scale	Best For
2mA	2 milliamps	General-purpose current measurement, LED testing
200uA	200 microamps	Moderate leakage currents, sensor characterisation
20uA	20 microamps	Low-current device testing, photodetector characterisation
2uA	2 microamps	Insulation resistance testing, low-leakage semiconductors
200nA	200 nanoamps	High-resistance material characterisation
20nA	20 nanoamps	Ultra-low leakage measurement, capacitor leakage
2nA	2 nanoamps	Picoammeter-grade measurement, thin-film characterisation
200pA	200 picoamps	Ultra-low current measurement, guard-ring structures

Choosing the correct range: Select the lowest range that can accommodate the expected current without over-ranging. A lower range provides finer resolution and lower noise. If the measured current exceeds the selected range, the display will indicate an over-range condition.

For best accuracy at very low currents (2nA and 200pA ranges), allow the instrument to warm up for a few minutes and use the Null/Zero function to cancel any offset before taking measurements.

7.2 Sample Rate

The sample rate determines how frequently the instrument acquires current readings. Select the sample rate using the **Sample Rate** control.

Sample Rate	Description
2Hz	Slowest rate. Provides the lowest noise and best accuracy. Ideal for stable, low-current measurements where speed is not critical.
10Hz	Balanced rate. Good compromise between noise performance and update speed. Suitable for most general-purpose measurements.
100Hz	Fastest rate. Best for capturing rapidly changing currents or for fast data logging. Higher noise floor than slower rates.

A slower sample rate allows the instrument to average over a longer integration period, reducing noise and improving measurement stability. Use 2Hz for the most demanding low-current measurements and 100Hz when you need to capture transient behaviour.

7.3 Null / Zero

The **Null** (also called **Zero**) function cancels residual offset current in the measurement path. This is essential for low-current measurements where offsets from cables, connectors, and the instrument's input circuitry can be significant.

To perform a null/zero:

1. Disconnect the device under test (or ensure zero current is flowing)
2. Allow the reading to stabilise
3. Click the **Null** button
4. The instrument captures the current offset and subtracts it from all subsequent readings

The null offset is stored for the currently selected range. If you change ranges, you may need to perform the null procedure again.

On the SPA120, each channel has its own independent Null function. Click the Null button in the Ch.1 or Ch.2 controls to null that channel separately.

Null/zero is particularly important when measuring on the 200pA, 2nA, and 20nA ranges, where input offset currents can be a significant fraction of the full-scale reading.

7.4 Rolling Averaging

Rolling averaging reduces noise by computing a moving average of multiple consecutive readings. Select the averaging factor using the **Averaging** control.

Averaging Factor	Effect
1x	No averaging. Each displayed reading is a single sample.

Averaging Factor	Effect
2x	Average of 2 consecutive readings
4x	Average of 4 consecutive readings
8x	Average of 8 consecutive readings
10x	Average of 10 consecutive readings
16x	Average of 16 consecutive readings
20x	Average of 20 consecutive readings
32x	Average of 32 consecutive readings
40x	Average of 40 consecutive readings
64x	Average of 64 consecutive readings. Maximum smoothing.

Trade-off: Higher averaging factors produce a smoother, more stable reading but increase the response time. With 64x averaging at 2Hz sample rate, the displayed reading reflects the average over 32 seconds of data. For fast-changing signals, use a lower averaging factor or no averaging (1x).

Typical usage:

- **1x to 4x** - Fast-changing signals, transient capture, initial setup
- **8x to 20x** - General-purpose measurement, moderate noise reduction
- **32x to 64x** - Stable DC measurements where maximum noise rejection is needed

8 Source Supply

The SPA includes a programmable bias voltage source for applying a controlled voltage to the device under test. The source supply controls are located in the left-hand ribbon of the main display.

8.1 Voltage Control

The source supply provides a continuously adjustable output from **0V to +40V** (or **-40V** when negative polarity is selected). Use the **+** and **-** buttons to step through preset voltage levels, or click the **centre** of the control to enter a custom voltage value directly via a dialog.

The voltage output is suitable for biasing semiconductors, driving sensors, characterising high-resistance materials, and measuring insulation resistance.

Always start with a low voltage when testing an unknown device and increase gradually to avoid damage.

8.2 Polarity

The source supply polarity determines whether a positive or negative voltage is applied to the output terminals. Click the **polarity toggle** to switch between Positive and Negative.

- **Positive:** Applies a positive bias voltage (0V to +40V)
- **Negative:** Applies a negative bias voltage (0V to -40V)

Select the polarity appropriate for your device under test and measurement configuration.

8.3 Source On/Off

The **Source** button enables or disables the bias voltage output.

- Click **Source** to toggle the voltage output on or off
- The Source button indicator in EPIC shows the current state
- The source will automatically disable if a communication error occurs
- On the SPA120, each channel has its own Source on/off control

Always ensure the device under test is properly connected before enabling the source output. The Source button must show OFF before connecting or disconnecting devices.

9 Graph Display

The main graph area plots measured current against time.

9.1 Main Graph

Current (Y-axis) is plotted against time (X-axis). The display updates continuously as new readings are acquired, showing stability, transient events, drift, and the effect of voltage changes on current.

9.2 Graph Title

Click the graph title at the top of the main graph to edit it. The title is saved with setup files.

9.3 Graph Key

Below the main graph, a colour-coded key identifies each channel's trace. Ch.1 is shown in yellow, Ch.2 in cyan. Toggle the key on or off from the **Graph Keys** button in the Advanced panel.

9.4 Sub-Graph Overview

Below the main graph, a **sub-graph overview** provides a miniature view of the entire recording. This is particularly useful for long recordings where only a portion of the data fits in the main graph view.

Key features of the sub-graph:

- Shows the complete recording at all times, giving context for where the current view sits within the full dataset
- **C and D markers** (vertical dashed lines with drag handles) indicate which portion of the recording is currently displayed in the main graph. These are labelled C and D to distinguish them from the main graph's A and B measurement cursors
- **Drag the C or D markers** to manually scroll through the recording. Dragging either marker moves the entire view window — C marks the left (start) edge and D marks the right (end) edge
- **Click within the sub-graph** to reposition the view

Auto-follow mode: When first started, the D marker automatically tracks the right edge of the incoming data so the main graph always shows the latest measurements. This is indicated by the **D marker turning green**. If you drag either C or D marker to browse earlier data, auto-follow disengages and the D marker returns to the normal grid colour. To re-engage auto-follow, simply drag the D marker back to the right edge of the sub-graph — it will snap to auto-follow mode and turn green again

Trace dimming: Click on a column of the sub-graph to dim that trace, useful for focusing on specific time regions or comparing before/after segments.

Splitter bar: The boundary between the main graph and sub-graph can be dragged vertically to adjust the relative sizes of each graph. Drag up to give more space to the sub-graph, or down to enlarge the main graph. The splitter position is saved with setups.

9.5 Timebase

The **Timebase** control in the left ribbon sets the horizontal time scale (seconds per division) for the main graph. Use the + and - buttons to zoom the time axis in or out. The timebase determines how much time is visible in the main graph window at once. Smaller timebase values show finer detail; larger values show more data.

9.6 Y-Axis Scale

The **Y-Scale** control sets the vertical current scale (amps per division) for the main graph. Use the **+** and **-** buttons to adjust the vertical zoom. The Y-axis scale determines how much current range is visible in the main graph.

9.7 Y-Axis Mode

Below the Y-Scale control, four mode buttons control how the Y-axis is positioned:

Mode	Description
AUTO	Automatically adjusts the Y-axis range to fit all visible data. Best for general use.
CENTRE	Centres the Y-axis around the current live reading, keeping the trace mid-screen.
FORCE+	Forces the Y-axis positive, placing zero at the bottom of the graph. Useful when measuring only positive currents.
FORCE-	Forces the Y-axis negative, placing zero at the top of the graph. Useful when measuring only negative currents.

9.8 Recording

Click the **Run** button to begin recording current measurements to the graph. The instrument will continuously acquire and display readings at the selected sample rate.

Click the **Stop** button to end the recording session. The session is automatically saved, and the graph is cleared ready for the next recording.

9.9 Pause and Resume

Click the **Pause** button to temporarily suspend recording without ending the session. The graph retains all data collected so far.

Click **Pause** again (or **Run**) to resume recording. New data is appended to the existing session. The boundary between paused and resumed segments is marked on the graph.

9.9.1 Pause Markers

In Local Time and UTC display modes, pause boundaries appear as a filled gap on the graph with a rotated text label showing the duration of the pause (e.g. "RECORDING PAUSED FOR 00:01:23.456").

9.9.2 Range Change Warning

If the measurement range is changed on either channel during a pause, a bright red warning is displayed alongside the pause duration text:

CH1 RANGE: 200 pA → 2 nA | CH2 RANGE: 20 nA → 200 nA

This alerts you that data either side of the pause was measured with different sensitivity, so readings across that boundary are not directly comparable. Warnings appear independently for each channel — only the channel(s) whose range actually changed will be shown.

10 Session Browser

The Session Browser provides access to previously recorded measurement sessions.

10.1 Browsing Sessions

Use the < > arrow buttons to step through saved session files. The session data is loaded and displayed on the graph, allowing you to review historical measurements.

10.2 Opening a Specific Session

Click the **centre** of the Session Browser panel (where the session name is displayed) to open a file chooser dialog. This lets you navigate directly to any session JSON file in the sessions folder without stepping through files one at a time.

10.3 Returning to Live Data

After browsing session files, start a new recording to return to live data display.

11 Results Display

The right-hand ribbon displays live numerical data from the instrument.

11.1 Live Current

The primary result displayed is the **live current** reading, shown in engineering notation with appropriate units (mA, uA, nA, or pA). This value updates continuously at the selected sample rate.

11.2 Max / Min Tracking

The results display tracks and shows:

- **Maximum (MAX+)** - The highest current reading observed since the last reset
- **Minimum (MAX-)** - The lowest (most negative) current reading observed since the last reset

Click on the **MAX+** display to reset the positive maximum to the current reading, or click on the **MAX-** display to reset the negative maximum independently. Each reset only affects the value you click on. If a tracked value exceeds 4 digits, the display shows **OVR** (over-range) instead of the numeric value.

Max/Min tracking identifies peak and trough values during a session.

11.3 Resistance Calculation

The SPA calculates and displays the resistance of the device under test using Ohm's law:

$$R = V / I$$

Where **V** is the applied bias voltage and **I** is the measured current. This provides a real-time resistance reading that updates as the voltage or current changes.

The resistance reading requires a non-zero bias voltage.

12 Cursors

The SPA provides a full cursor measurement system for making precise measurements on recorded graph data.

12.1 X Cursors

Click the **X Cursor** button on the toolbar to enable horizontal (time) cursors. Two vertical cursor lines (A and B) appear on the main graph.

- **Drag** cursor A or B directly on the graph to position them
- Use the +/- fine-adjust buttons in the cursor controls to nudge cursors by small increments
- Cursor values are displayed in absolute time (seconds from the start of recording), regardless of the current view position

12.2 Y Cursors

Click the **Y Cursor** button on the toolbar to enable vertical (current) cursors. Two horizontal cursor lines (A and B) appear on the main graph.

- **Drag** cursor A or B directly on the graph to position them
- Use the +/- fine-adjust buttons to nudge cursors by small increments
- Cursor values are displayed with the correct current units (A, with appropriate SI prefix)

12.3 Slope Measurement

Click the **Slope** button to enable slope measurement between the cursor positions:

$$\text{Slope} = (XB - XA) / (YB - YA)$$

12.4 Cursor Results

Toggle the cursor results table using the **Cursor Measurements** button. When visible, the table appears in the right-hand ribbon and displays:

Measurement	Description
XA	Time position of cursor A
XB	Time position of cursor B
XB-XA	Time difference between cursors (delta time)
YA	Current value at cursor A
YB	Current value at cursor B
YB-YA	Current difference between cursors (delta current)
Slope	Rate of change (when slope measurement is enabled)

Activating **Slope** mode automatically enables the cursor results table.

13 Dual Channel Features

The SPA120 provides two independent measurement channels, referred to as **Ch.1** and **Ch.2**. Each channel has its own current measurement input, bias voltage source, and associated controls.

13.1 Channel Operation

- **Ch.1:** The primary measurement channel. All controls in the main left ribbon apply to Ch.1.
- **Ch.2:** The secondary measurement channel. Ch.2 controls appear in a dedicated section below the Ch.1 controls.

Each channel operates independently with its own:

- Current measurement range
- Bias voltage setting and polarity
- Null/zero offset
- Source on/off control

Both channels share the same sample rate and rolling averaging settings.

13.2 Dual Channel Display

When using the SPA120, the graph display shows both channels simultaneously. Ch.1 is plotted in yellow and Ch.2 in cyan, allowing direct visual comparison. The bottom information text also shows settings for both channels in their respective trace colours.

The results display shows live readings for both channels, including individual current, max/min, and resistance values.

13.3 Typical Applications for Dual Channel Operation

- **Comparative measurement:** Measure two similar devices side-by-side under identical conditions to identify differences
 - **Differential measurement:** Monitor current at two points in the same circuit simultaneously
 - **Before/after testing:** Use one channel as a reference and the other for the device under test
 - **Matched pair selection:** Compare devices to find closely matched pairs for precision circuits
-

14 CSV Export

The CSV export feature saves measurement data in comma-separated values format, suitable for opening in spreadsheet applications (such as Microsoft Excel or LibreOffice Calc) or importing into analysis software.

14.1 Saving CSV Data

Click the **CSV** button on the toolbar to save the current recorded data. A file save dialog will appear allowing you to choose the filename and location.

14.2 CSV File Format

The SPA CSV file contains **9 columns** with one row per recorded sample:

Column	Description
Sample	Sample number (sequential from 0)
Time (s)	Time in seconds from the start of recording
UTC Time	UTC timestamp for each sample (YYYY-MM-DD HH:MM:SS format)
Ch1 Current (A)	Channel 1 measured current in Amps
Ch2 Current (A)	Channel 2 measured current in Amps
Input Range Ch1	Channel 1 input range setting text
Input Range Ch2	Channel 2 input range setting text
Source Ch1 (V)	Channel 1 source voltage setting
Source Ch2 (V)	Channel 2 source voltage setting

14.3 Configuration Options

Configure CSV export in the **Settings** window under the **File Export** tab:

Setting	Description
Save Location	Folder where CSV files are saved
Auto-Name	Automatically generate sequential filenames
Prefix	Text prefix for auto-named files

14.4 Auto-Naming

When auto-naming is enabled, files are saved with a sequential counter:

[Prefix]_001.csv, [Prefix]_002.csv, [Prefix]_003.csv...

The counter can be reset or preset to a specific value in Settings.

15 PNG Capture

The PNG capture feature saves the current graph display as a PNG image file.

15.1 Saving a PNG

Click the **PNG** button on the toolbar to save the current display as a PNG image.

15.2 Configuration Options

Configure PNG capture in the **Settings** window under the **File Export** tab:

Setting	Description
Save Location	Folder where PNG files are saved
Auto-Name	Automatically generate sequential filenames
Prefix	Text prefix for auto-named files
Include Sub-Graph	Include the sub-graph area in the capture (checked by default)
Include Top Ribbon	Include the top ribbon (toolbar) in the capture
Include Left Ribbon	Include the left ribbon (controls) in the capture
Include Right Ribbon	Include the right ribbon (results) in the capture
Use CSV Name	Match the PNG filename to the CSV filename

The capture always includes the main graph area. Use the checkboxes to extend the capture to include surrounding ribbons. By default only the sub-graph is included.

15.3 Auto-Naming

When auto-naming is enabled, files are saved with a sequential counter:

[Prefix]_001.png, [Prefix]_002.png, [Prefix]_003.png...

The counter can be reset or preset to a specific value in Settings.

16 Metadata Overlay

The Metadata Overlay displays key settings as a semi-transparent table on the graph. Toggle it using the **Info** button on the top ribbon or via the Advanced panel. Each click cycles through four corner positions before turning off.

16.1 Displayed Information

Field	Description
Ch.1 Range	Channel 1 measurement range (e.g. 200 pA, 2 nA, 20 nA)
Ch.1 Null	Channel 1 null offset, shown only when Null is active
Ch.1 Source	Channel 1 bias supply voltage; displays "0 V" when the source is off
Ch.2 Range	Channel 2 measurement range
Ch.2 Null	Channel 2 null offset, shown only when Null is active
Ch.2 Source	Channel 2 bias supply voltage; displays "0 V" when the source is off
Sample Rate	Current sample rate (2 Hz, 10 Hz, or 100 Hz)
Averaging	Rolling average multiplier (e.g. 10x)

The Null rows only appear when Null is enabled on the respective channel. Null values are displayed in engineering notation with 5 significant digits (e.g. "20.000 nA" or "1.9546 pA"). Channel-specific information is grouped together: Range, then Null (if active), then Source for each channel.

16.2 Draw Order

By default the INFO box is drawn in front of the traces, which may obscure parts of the measurement data. To draw the INFO box behind the traces instead, go to **Settings > Graph Tab** and change the **INFO Box Draw Order** to "Behind Traces". This allows traces to be drawn over the INFO box, keeping the measurement data fully visible.

17 Data Streaming

Data Streaming writes measurement data continuously to external files and services as it is acquired. Unlike the standard CSV export, there is no row limit.

Data Streaming is configured in the **Settings** window under the **Data Streaming** tab.

17.1 Overview

When one or more streaming outputs are enabled, every measurement sample is buffered internally and periodically dispatched to the enabled outputs. The dispatch interval is configurable and shared across all outputs — it can be set in **samples** (e.g. every 100 samples) or **seconds** (e.g. every 10 seconds).

Three output types are available:

Output	Description	Status
CSV Write	Writes batches of samples to timestamped CSV files in a chosen directory	Available
HTTP POST	Sends batches of samples as JSON to a remote URL	Available
MQTT Send	Publishes sample data to an MQTT broker	Future

Each output is independently enabled — you can use any combination simultaneously.

17.2 CSV Streaming

When CSV Write is enabled, EPIC creates a new CSV file for each batch of samples. Files are written to the configured directory (default: `stream\`) and named using the UTC timestamp of the first sample in the batch:

`stream_20260308_142345_123.csv`

17.2.1 CSV File Format

Each CSV file contains a header row followed by one row per sample. The SPA120 includes both channel 1 and channel 2 columns:

Column	Description
Sample	Monotonic sample counter (never resets during a session)
Instrument	Instrument and serial number, e.g. SPA120:1X1X1X
UTC Time	ISO 8601 timestamp with milliseconds, e.g. 2026-03-08T14:23:45.123Z
Ch1 Value (A)	Channel 1 measured current in Amps (scientific notation, full precision)
Ch2 Value (A)	Channel 2 measured current in Amps (scientific notation, full precision)
Ch1 Range (A)	Channel 1 input range in Amps (scientific notation)
Ch1 Source (V)	Channel 1 source supply voltage (scientific notation)
Ch1 Source Enable	Channel 1 source on/off state (1 = On, 0 = Off)
Ch2 Source (V)	Channel 2 source supply voltage (scientific notation)
Ch2 Source Enable	Channel 2 source on/off state (1 = On, 0 = Off)
Ch1 Null Offset (A)	Channel 1 null/zero offset value in Amps (scientific notation)
Ch1 Null Enable	Channel 1 null engaged (1 = On, 0 = Off)
Ch2 Null Offset (A)	Channel 2 null/zero offset value in Amps (scientific notation)

Column	Description
Ch2 Null Enable	Channel 2 null engaged (1 = On, 0 = Off)
Ch1 Averaging	Channel 1 rolling averager count (e.g. 1, 2, 4, 8, 10, 16, 20, 32)
Ch2 Averaging	Channel 2 rolling averager count
Ch1 Short Enable	Channel 1 input short engaged (1 = On, 0 = Off)
Ch2 Short Enable	Channel 2 input short engaged (1 = On, 0 = Off)

All current and voltage values are output in scientific notation at full IEEE 754 double precision (15 significant digits), e.g. 1.85123456789012e-09. This avoids any rounding that could affect post-processing.

17.2.2 CSV Streaming vs Standard CSV Export

Feature	Standard CSV Export	CSV Streaming
Trigger	Manual (click CSV button)	Automatic (continuous)
Row limit	Limited by in-memory buffer	Unlimited — writes directly to disk
File count	Single file	One file per batch
Precision	Fixed decimal	Full scientific notation
Timestamps	Relative seconds + basic UTC	ISO 8601 with milliseconds

17.3 HTTP POST

When HTTP POST is enabled, EPIC sends each batch of samples as a JSON payload to the configured URL. This enables real-time data collection by remote servers, cloud dashboards, or automation systems.

17.3.1 JSON Payload Format

```
{
  "version": "V26006",
  "instrument": "SPA120:1X1X1X",
  "pushTime": "2026-03-08 14:23:45",
  "sampleCount": 100,
  "status": {
    "recording": 1
  },
  "samples": [
    {
      "sample": 1,
      "instrument": "SPA120:1X1X1X",
      "utcTime": "2026-03-08T14:23:35.767Z",
```

```
"ch1Value": -1.79719729476114e-12,  
"ch2Value": 3.45678901234567e-11,  
"ch1Range": 0.002,  
"ch1Source": 1.5,  
"ch1SourceEnable": 1,  
"ch2Source": 0.0,  
"ch2SourceEnable": 0,  
"ch1NullOffset": 0.0,  
"ch1NullEnable": 0,  
"ch2NullOffset": 0.0,  
"ch2NullEnable": 0,  
"ch1Averaging": 1,  
"ch2Averaging": 1,  
"ch1ShortEnable": 0,  
"ch2ShortEnable": 0  
}  
]  
}
```

17.3.2 Authentication

If your endpoint requires authentication, enter a **Bearer token** in the Auth Token field. EPIC will include it in the HTTP Authorization header:

Authorization: Bearer <your-token>

17.3.3 HTTP POST runs on a background thread

The HTTP request is performed on a separate thread so that the main EPIC display and measurement loop are never blocked, even if the server is slow to respond or unreachable.

17.4 Testing HTTP POST with webhook.site

The easiest way to verify that HTTP POST is working is to use **webhook.site**, a free online service that gives you a temporary URL and displays all incoming requests in real time.

17.4.1 Step-by-step

1. Open your web browser and go to **https://webhook.site**
2. The site will display a unique URL at the top of the page, for example:

https://webhook.site/abc12345-6789-def0-1234-abcdef567890
3. Copy this URL to the clipboard
4. In EPIC, open **Settings** and select the **Data Streaming** tab
5. Tick the **HTTP POST** checkbox under Output Enables
6. Paste the webhook.site URL into the **URL** field
7. Set the streaming interval as desired (e.g. 100 samples or 10 seconds)

8. Close the Settings window
9. Start recording on the SPA120 — click **Run**
10. Switch back to your web browser — you should see JSON payloads appearing on the web-hook.site page within a few seconds

Each incoming request will show:

- The full JSON body containing your measurement samples (including both Ch1 and Ch2 data)
- HTTP headers including Content-Type: application/json
- The timestamp of each request

webhook.site URLs are temporary and the free tier typically accepts only 50 requests before requiring a new URL. Use a low dispatch interval (e.g. 1000 samples) or short test runs to stay within this limit. For permanent data collection, set up your own HTTP endpoint or use a cloud service.

17.4.2 Alternative test endpoints

- **Postman Echo** — <https://postman-echo.com/post> — echoes your payload back as a JSON response
- **Local Python server** — for offline testing, save this as `postserver.py` and run with `python postserver.py`:

```
from http.server import HTTPServer, BaseHTTPRequestHandler
import json

class Handler(BaseHTTPRequestHandler):
    def do_POST(self):
        length = int(self.headers.get('Content-Length', 0))
        body = self.rfile.read(length)
        print(json.dumps(json.loads(body), indent=2))
        self.send_response(200)
        self.end_headers()
        self.wfile.write(b'OK')

HTTPServer(('', 8080), Handler).serve_forever()
```

Then set the EPIC HTTP POST URL to `http://localhost:8080`.

17.5 Streaming Interval

The streaming interval controls how often buffered samples are dispatched to all enabled outputs. It is a single shared setting — CSV and HTTP POST both use the same interval.

Two modes are available:

Mode	Description	Example
Samples	Dispatch after a fixed number of samples have been collected	100 samples at 10 Hz = every 10 seconds
Seconds	Dispatch after a fixed time period has elapsed	Every 30 seconds, regardless of sample rate

In **Samples mode**, the dispatch is triggered immediately when the sample count is reached. In **Seconds mode**, the dispatch is checked once per main loop iteration (typically 20–50 times per second) so the actual timing is accurate to within ~50 ms.

Available presets:

Samples mode	Seconds mode
50	5
100	10
200	30
500	60
1000	300

18 Gang Range and Gang Source (SPA120 Only)

The **Gang Range** and **Gang Source** features link Channel 2 settings to Channel 1, so that changes to Channel 1 automatically propagate to Channel 2.

These are toggle buttons available in the **Miscellaneous** section of the smorgasbord panel, and on the top ribbon when enabled.

18.1 Gang Range

When **Gang Range** is active:

- Changing the Ch.1 measurement range automatically sets Ch.2 to the same range
- Toggling Ch.1 auto-range mode also toggles Ch.2
- The Ch.2 range controls are dimmed and locked — they cannot be adjusted independently
- Auto-ranging on Ch.1 also drives Ch.2 to the same range

18.2 Gang Source

When **Gang Source** is active:

- Changing the Ch.1 source voltage automatically sets Ch.2 to the same voltage
- Toggling Ch.1 source polarity also toggles Ch.2
- Switching Ch.1 source on/off also switches Ch.2
- The Ch.2 source controls are dimmed and locked — they cannot be adjusted independently

18.3 Disabling Gang

Click the Gang Range or Gang Source button again to disable ganging. Once disabled, the Ch.2 controls return to normal and can be adjusted independently. The Ch.2 settings remain at whatever values they were synced to — they are not reverted.

Gang features are useful when you want both channels to measure under identical conditions, for example when comparing two devices or monitoring two points in the same circuit with matched settings.

19 Save / Load Setups

19.1 Saving a Setup

1. Click the **Save Setup** button (in the Advanced panel or top bar)
2. Choose a filename and location in the save dialog
3. The current configuration is saved as a JSON file

What is saved:

- Current measurement range (both channels)
- Bias voltage and polarity settings (both channels)
- Source on/off state
- Sample rate and rolling averaging factor
- Null/zero offset values
- Export settings (CSV/PNG paths, auto-naming, prefixes)
- Display settings (graph title, trace colours)
- Graph configuration (sub-graph height, time display mode, UTC offset)
- Cursor positions and enable states
- Both Ch.1 and Ch.2 channel configurations

19.2 Loading a Setup

1. Click the **Load Setup** button
2. Select a previously saved setup file (.json)
3. All settings are restored from the file

The setup file includes the instrument type. A setup saved for the SPA120 cannot be loaded on a different instrument type.

20 Settings

The **Settings** window provides access to all application configuration options. Open it by clicking the **Settings** button.

When an SPA instrument is selected, the settings window displays only the tabs relevant to SPA operation. CTL503-specific tabs (Test/Graph Mode, Collector Supply, Step Generator, Sweep/Live Values, Key/Measurement, Sweep Settings) are automatically hidden.

The available settings tabs for SPA instruments are:

20.1 Language Tab

Select the display language for the software interface. EPIC supports a wide range of languages including English, French, German, Spanish, Italian, Portuguese, Japanese, Chinese, Korean, Arabic, Hindi, Russian, and many more.

20.2 Instrument Tab

Select which instrument the software should communicate with. The instrument selection is organised by instrument family:

- **Row 1:** CTL503 (Curve Tracer) - with COM port displayed
- **Row 2:** SPA100, SPA120 (Source Picoammeters) - with COM ports displayed

The currently active instrument is highlighted in green. Each instrument button shows the assigned COM port number below it.

20.3 Comms Tab

Configure communication settings:

- COM port selection and scanning
- Connection method (auto-detect or manual)
- Manual COM port entry

20.4 File Export Tab

Configure CSV and PNG export settings:

- File save locations
- Auto-naming options and prefixes
- PNG capture area (full window or graph only)
- Sequential counter management

20.5 Graph Tab

Configure graph display options including trace colours and display preferences:

- **Display Decimation Algorithm** - Choose between Min/Max (preserves peaks) or LTTB (preserves shape) for trace rendering
- **INFO Box Draw Order** - Controls whether the Metadata Overlay (INFO box) is drawn in front of or behind the measured traces. When set to "Behind Traces", the traces are drawn on top of the INFO box, making them more visible. The default is "In Front of Traces"

20.6 Cursors Tab

Configure cursor behaviour and display options.

20.7 Calibration Tab

Instrument calibration management. The SPA120 is factory-calibrated and adjusted during production testing at Electron+ facilities. Recalibration can be performed via the EPIC calibration utility. The calibration data is stored in `settings/SPA120_cal.JSON`.

20.8 Colours Tab

Customise the display colours for traces, backgrounds, and UI elements. Trace colours can be independently set using RGB values with real-time preview.

20.9 Setups Tab

Manage instrument setup files, including save/load configuration presets.

20.10 Data Streaming Tab

Configure continuous data output to external files and services. Enable or disable each output independently (CSV Write, HTTP POST, MQTT Send), set the shared streaming interval in samples or seconds, configure the CSV output directory, and enter the HTTP POST URL and optional authentication token. A live status display shows the total samples streamed, last CSV file written, and most recent HTTP response code.

21 Calibration

The SPA120 is factory-calibrated and adjusted during production testing at Electron+ facilities. If recalibration is required:

1. Open **Settings** and navigate to the **Calibration** tab
2. Follow the on-screen calibration procedure
3. The calibration data is stored in `settings/SPA120_cal.JSON`

Calibration ensures accuracy across all eight current measurement ranges and the voltage source output.

22 Advanced Panel (Smorgasbord)

The Advanced panel (also known as the Smorgasbord) allows you to toggle the visibility of various toolbar buttons and controls. Click the **Advanced** button to open the panel.

Each button on the smorgasbord panel has a small **blue ring** indicator in its top-left corner. A filled blue circle means the button is currently enabled and visible in the main ribbon. A hollow blue ring means the button is hidden. Click the blue ring to toggle a button's visibility without changing its function — this lets you keep the main ribbon uncluttered by hiding controls you don't use frequently.

22.1 Export Functions

- **PNG** - Show/hide the PNG capture button
- **CSV** - Show/hide the CSV export button

22.2 Cursor/Measurement Functions

- **X Cursors** - Show/hide X-axis (time) cursors
- **Y Cursors** - Show/hide Y-axis (current) cursors
- **Slope** - Show/hide slope measurement
- **Cursor Measurements** - Show/hide the cursor results table

22.3 Graph Functions

- **Graph Keys** - Show/hide the colour-coded channel key below the graph (see Graph Key section)
- **Metadata Overlay** - Toggle the metadata overlay on the graph area (see Metadata Overlay section)

22.4 Miscellaneous

- **Save Setup** - Save current configuration
- **Load Setup** - Restore saved configuration
- **Gang Range** - Link Ch.2 range to Ch.1 (see Gang Range and Gang Source section)
- **Gang Source** - Link Ch.2 source to Ch.1 (see Gang Range and Gang Source section)

This is useful for simplifying the interface by hiding controls you don't regularly use, or for maximising the graph display area.

23 Safety Information

The SPA120 can generate voltages up to 40V at the output terminals. Always observe the following safety precautions:

- **Never touch** the output terminals while the source is enabled (Source button shows ON)
 - **Always disable the source** before connecting or disconnecting devices from the input terminals
 - Use appropriate **voltage settings** to protect devices under test
 - Ensure proper **grounding** of the instrument and test setup
 - Do **not** operate the instrument with damaged cables or connectors
 - Keep the instrument in a **dry environment** away from liquids
 - Use **shielded cables** for ultra-low current measurements to minimise noise pickup
 - **Guard connections** should be used where possible for measurements below 2nA to reduce leakage paths
-

24 Troubleshooting

24.1 Connection Problems

Symptom: Cannot connect to the instrument

- Check USB cable is firmly connected at both ends
- Verify the GREEN LED on the SPA is lit
- Check that the correct COM port is selected in Settings
- Try scanning for COM ports again
- Restart the instrument and software
- Check Windows Device Manager for the COM port

Symptom: Connection drops intermittently

-
- Use a shorter USB cable
 - Avoid USB hubs - connect directly to the computer
 - Check for electromagnetic interference near the USB cable

24.2 Measurement Problems

Symptom: Readings are noisy or unstable

- Select a slower sample rate (2Hz) for lower noise
- Increase the rolling averaging factor
- Use the Null/Zero function to cancel offsets
- Use shielded cables and keep leads short
- Ensure the test environment is free from vibration and air currents (important for ultra-low current measurements)
- Allow the instrument to warm up for a few minutes before taking critical measurements

Symptom: Readings are zero or very small

- Verify that the source output is enabled (Source button shows ON in EPIC)
- Check that the bias voltage is set to a non-zero value
- Verify that the device under test is properly connected
- Ensure the correct polarity is selected for your device
- Try a more sensitive (lower) current range

Symptom: Over-range indication (OVR displayed)

- Switch to a higher current range
- Reduce the bias voltage if the current is too high
- Check for short circuits in the test setup
- If MAX+ or MAX- shows OVR, this indicates the tracked peak value exceeded 4 digits; click the individual display to reset its tracking

Symptom: Large offset current on sensitive ranges

- Perform a Null/Zero calibration with no device connected
- Allow the instrument to warm up before nulling
- Check for contamination or moisture on connectors and cables
- Use guarded connections to minimise leakage paths

24.3 Display Issues

Symptom: Graph is not updating

- Verify that recording is active (Run button pressed)
- Check that the instrument is connected (GREEN LED lit)
- Ensure the correct instrument type is selected in Settings

Symptom: Sub-graph not visible

- Check that the splitter bar has not been dragged fully to the bottom
- Drag the splitter bar upward to reveal the sub-graph overview

Symptom: Cursors not responding to clicks or drags

- Ensure X or Y cursors are enabled via the toolbar buttons
- Click directly on the cursor line to initiate a drag
- Use the +/- fine-adjust buttons as an alternative to dragging

24.4 File Export Issues

Symptom: Cannot save CSV or PNG files

- Check that the save location exists and is writable
- Verify there is sufficient disk space
- Check file permissions on the save directory
- Try changing the save location in Settings

24.5 Data Streaming Issues

Symptom: CSV streaming files are not being created

- Check that **CSV Write** is ticked in Settings → Data Streaming → Output Enables
- Verify the streaming directory exists and is writable
- Confirm that recording is active (click **Run**) — streaming only outputs data while the instrument is acquiring samples
- Check the **Status** section at the bottom of the Data Streaming tab — the “Samples streamed” counter should be increasing

Symptom: HTTP POST is not sending data

- Check that **HTTP POST** is ticked in Settings → Data Streaming → Output Enables
- Verify the URL is correct and reachable from your computer
- Check the **HTTP last status** field in the Status section — a value of 200 indicates success
- If the status shows an error message, check your network connection and firewall settings
- Ensure the URL begins with http:// or https://
- If using an Auth Token, verify it is correct and has not expired

Symptom: Data appears in CSV files but not in HTTP POST (or vice versa)

- Each output is independently enabled — check that both are ticked under Output Enables
 - Both outputs share the same streaming interval — data is dispatched to all enabled outputs simultaneously
-

25 Support & Contact

If you need further assistance:

- **Email:** support@electron.plus
- **Website:** www.electron.plus

-
- **Manuals:** Latest manuals are available for download from the Electron+ website under the Manuals section

Electron+ provides free software updates for all instruments. Check the website regularly for the latest version of EPIC.